

HIGH PERFORMANCE COMPUTING FOR SCIENCE AND ENGINEERING

Harvard University
CS 2050

Lecture 2
January 29, 2026



High Performance Computing for Science and Engineering

Important Links

- [Canvas site.](#)
- [Compute cluster.](#)
- [GitHub page.](#)



High Performance Computing for Science and Engineering

CS 2050 and CSCI E-205

Instructor: Chuck Witt (wcw@seas.harvard.edu)

As manufacturing processes approach the physical limits of transistor density, efficient code must exploit parallelism to scale with available computing resources. Scientific software developers must therefore adopt a “think parallel” mindset to solve complex problems across academia, industry, and society. This course introduces parallel programming and its relationship to computer architectures, with an emphasis on high performance computing. Students will develop experience with programming models such as OpenMP, MPI, and CUDA, applying these techniques in homework and a term project.

Lectures

Week 1 - Introduction



Lecture 1 - Tues Jan 27

- Course overview
- Computing trends (Moore's Law, Flynn's taxonomy)

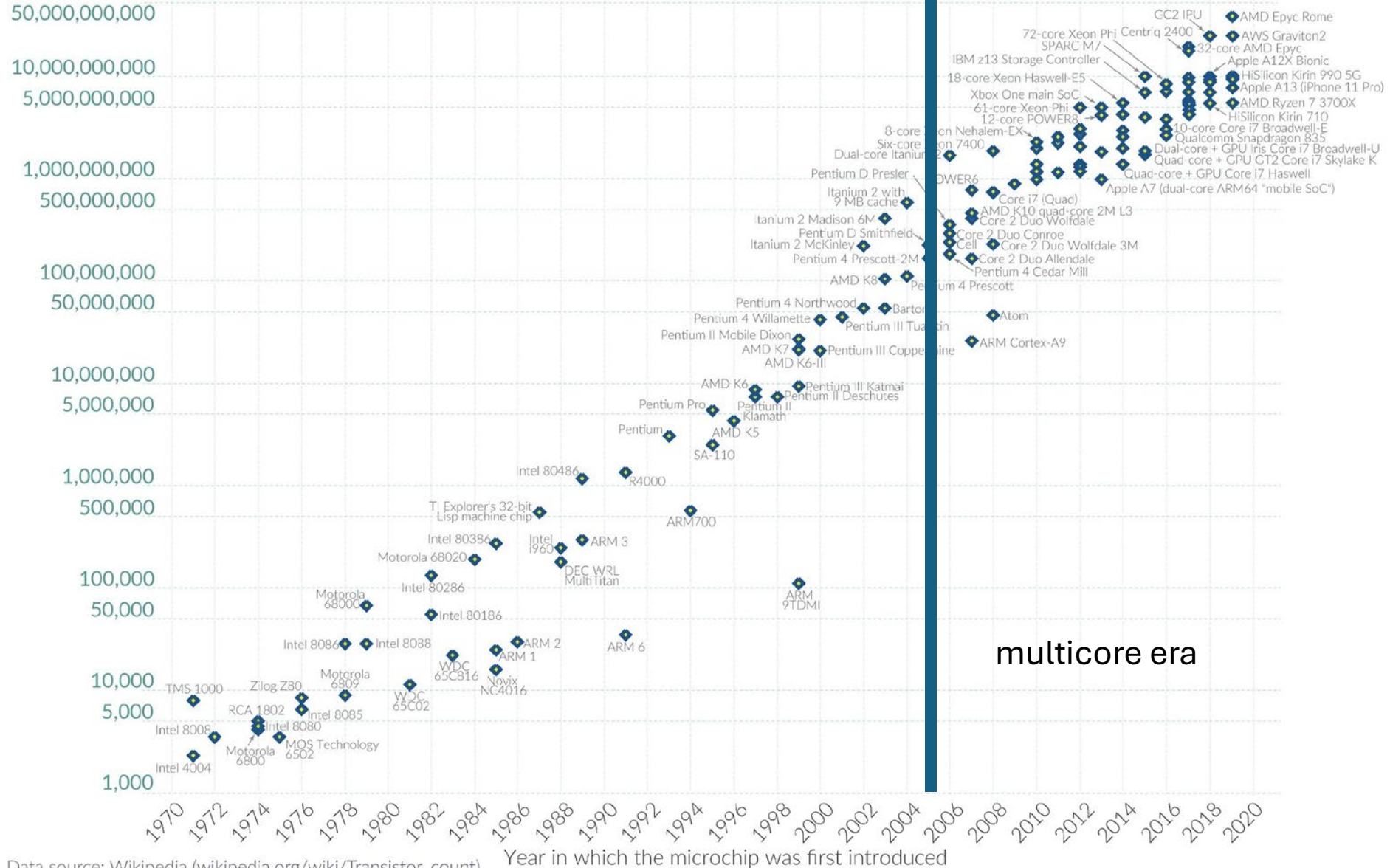
Lecture 2 - Thur Jan 29

- Using the compute cluster

Moore's Law: The number of transistors on microchips doubles every two years

Moore's law describes the empirical regularity that the number of transistors on integrated circuits doubles approximately every two years. This advancement is important for other aspects of technological progress in computing – such as processing speed or the price of computers.

Transistor count



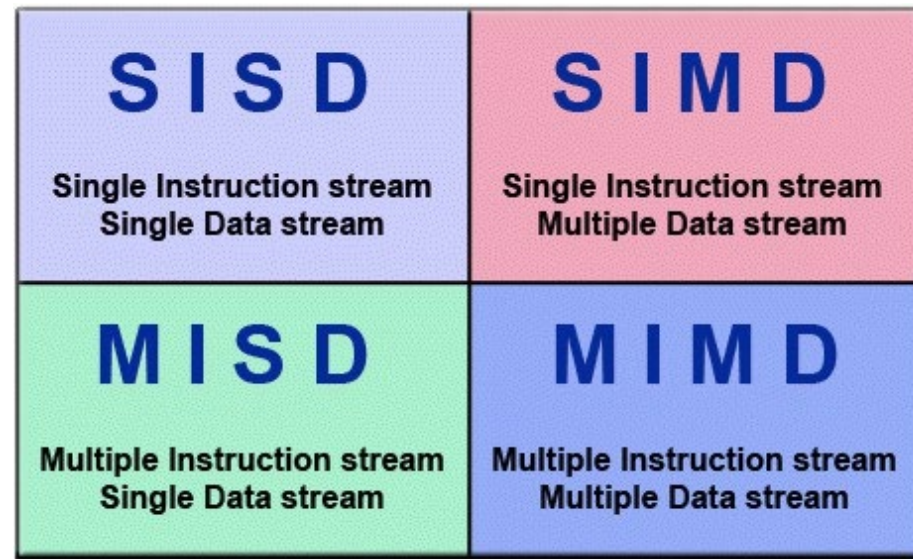
Data source: Wikipedia (wikipedia.org/wiki/Transistor_count)

OurWorldinData.org – Research and data to make progress against the world's largest problems.

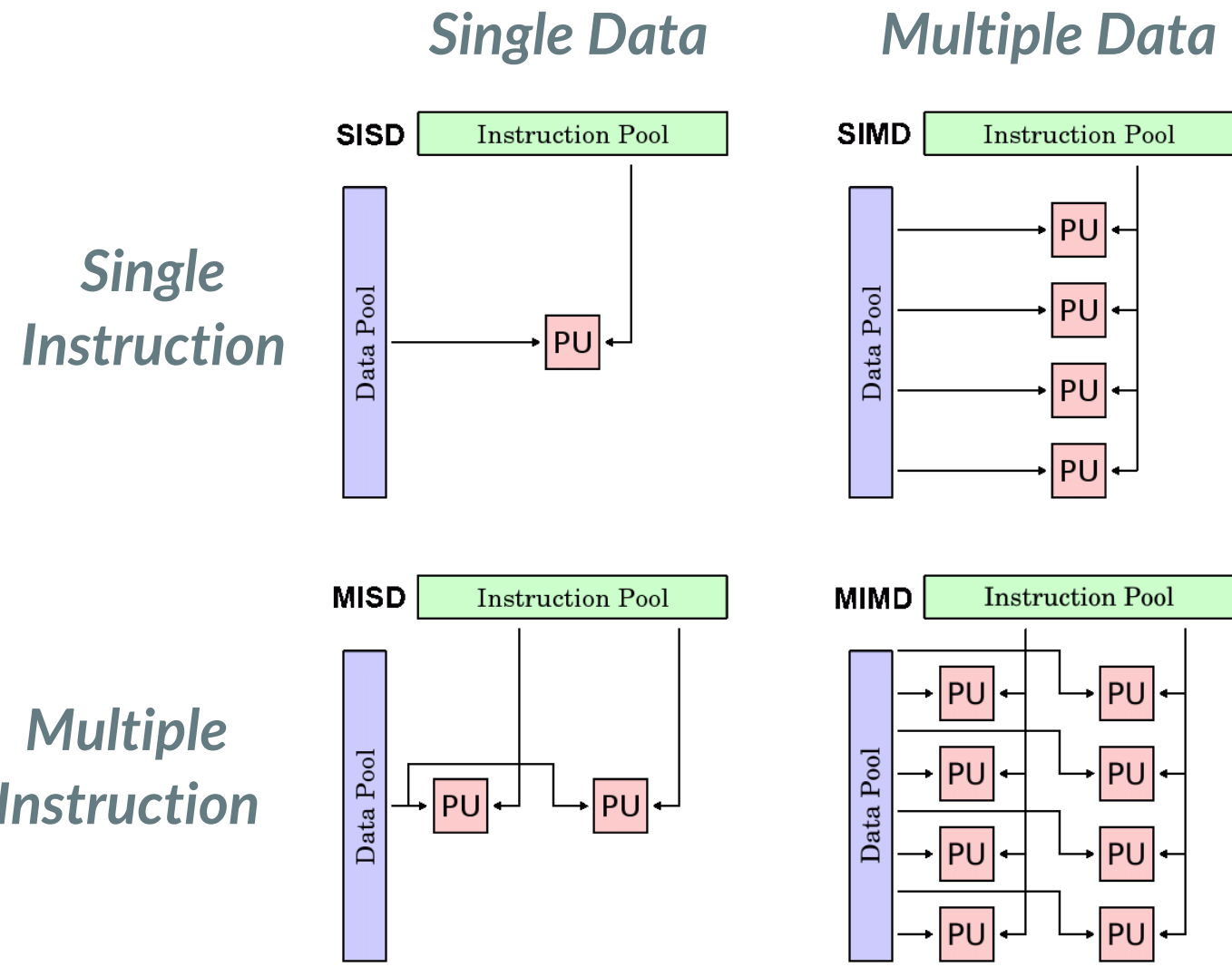
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FLYNN'S TAXONOMY

- Widely used classification of parallel computers
- Two independent dimensions: Instruction Stream and Data Stream
- Two possible states for each dimension: Single or Multiple.



FLYNN'S TAXONOMY





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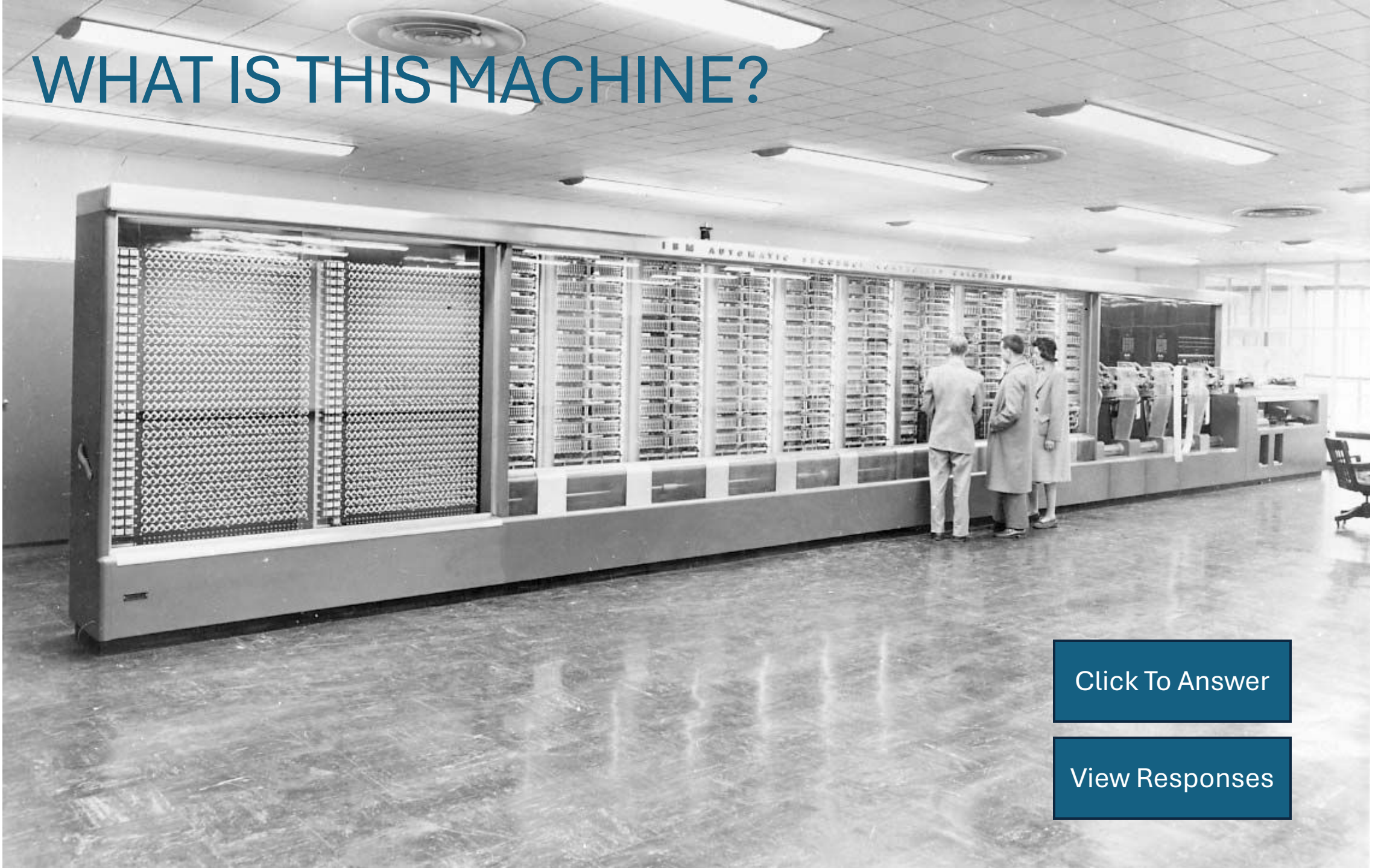
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WHAT IS THIS MACHINE?



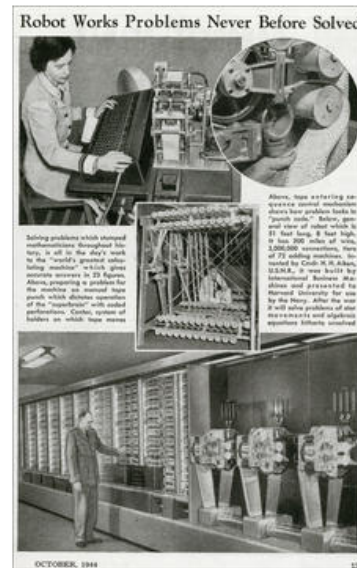
[Click To Answer](#)

[View Responses](#)

Harvard IBM Mark I – About

[About](#) | [Use](#) | [Crew](#) | [Function](#) | [Manual](#) | [Language](#) | [Futurism](#) | [Video](#)

About the Mark I



Mark I was designed in 1937 by Harvard graduate student Howard H. Aiken to solve advanced mathematical physics problems encountered in his research. Aiken's ambitious proposal envisioned the use of modified, commercially-available technologies coordinated by a central control system.

Supported by Harvard faculty in the division that is today the School of Engineering and Applied Sciences, Aiken discussed his idea with several manufacturers, eventually finding interest at IBM, a company that specialized in calculating machines and punch card systems. Using company components, IBM engineers in Endicott, NY developed the machine's working systems and directed its construction over five years. During that period America entered World War II. When Mark I was finally delivered to Harvard in 1944, it was operated by the U.S. Navy Bureau of Ships for military purposes, solving mathematical problems that until then required large teams of human "computers."

Mark I was in operation between 1944 and 1959, at which point sections from each of its components were taken to IBM and the Smithsonian Institution, leaving the smaller version seen here. The original Mark I was about twice the current length.

<https://chsi.harvard.edu/harvard-ibm-mark-1-about>
original image: <https://www.britannica.com/technology/Harvard-Mark-I>

WHO IS THIS?

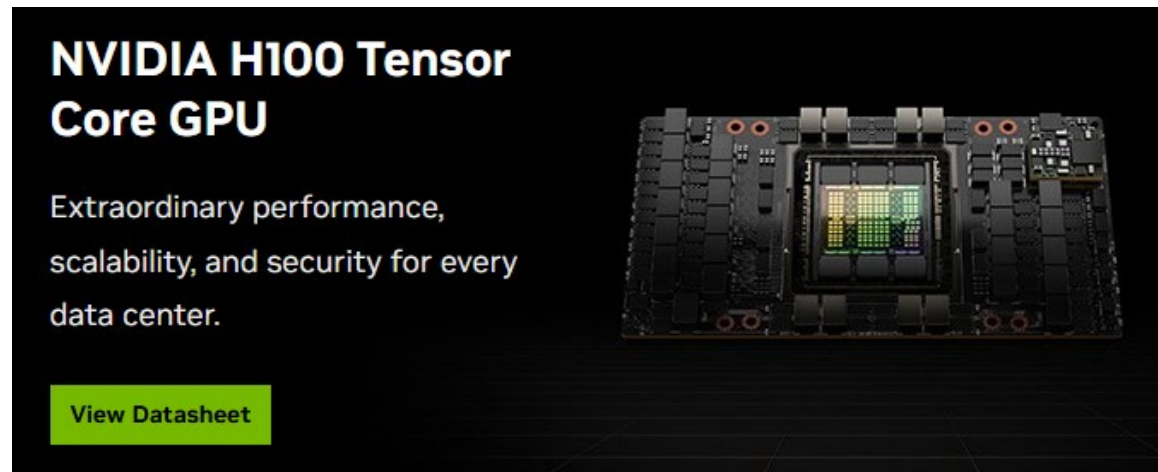


[Click To Answer](#)

[View Responses](#)

GRACE HOPPER

- “One of the first three computer ‘programmers,’ Hopper was responsible for programming the Mark I and punching machine instructions onto tape. She also wrote the 561-page user manual for the Mark I.”
- First “to speak of ‘debugging’ a computer”



ARE YOU COMFORTABLE USING THE LINUX COMMAND LINE?

[Click To Answer](#)

ARE YOU COMFORTABLE USING BASH SCRIPTS?

[Click To Answer](#)

ARE YOU COMFORTABLE USING C++?

[Click To Answer](#)

ARE YOU COMFORTABLE USING PYTHON?

[Click To Answer](#)

ARE YOU COMFORTABLE USING GIT AND GITHUB?

[Click To Answer](#)

ARE YOU COMFORTABLE USING CMAKE?

[Click To Answer](#)

ARE YOU COMFORTABLE USING SLURM?

[Click To Answer](#)

ARE YOU COMFORTABLE USING ARCHIBALD?

[Click To Answer](#)

ARE YOU COMFORTABLE USING VS CODE?

[Click To Answer](#)

GOAL FOR TODAY

- To connect to the compute cluster!

... and (more importantly) to identify problems quickly so that we can fix them over the next few days

Compute Cluster

Our compute cluster is a shared, finite resource, meaning submitted job(s) might not execute immediately and could be delayed by several hours or more. For this reason, students are **strongly encouraged** to complete assignments as early as feasible, and extensions will not be granted simply because the cluster was busy leading up to a deadline.



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OnDemand provides an integrated, single access point for all of your HPC resources.

Latest News for HUIT Open OnDemand

Updates to HUIT OOD configuration

The HUIT OOD configuration will be updated on January 9, 2025, at 1pm. Downtime and interruption of jobs is expected until 5pm. This update is being applied in order to prepare the infrastructure for the Spring 2025 term, and includes increased compute resources and updated base compute instances.

Fall 2024 End of Term

Access to HUIT Open OnDemand ends on **December 31st** for all Fall 2024 courses. Please ensure you download all your files and data before this date, as they will be permanently deleted. If you need continued access for your course, please contact ithelp@harvard.edu as soon as possible.

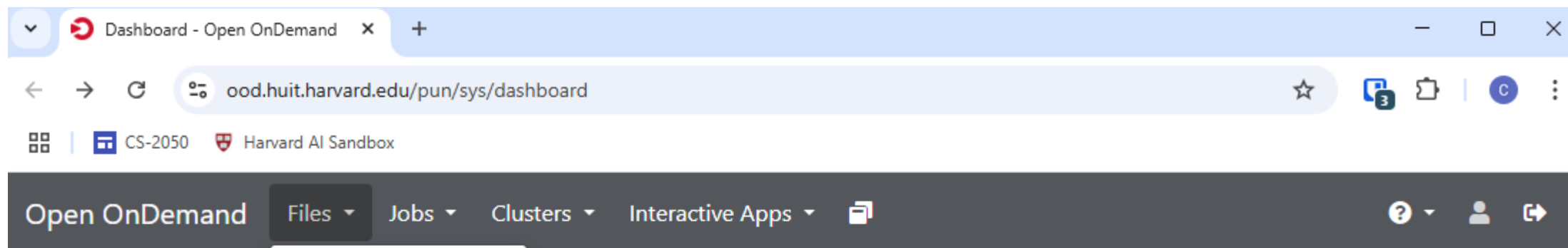
Where to go for help

For a basic overview of the HUIT implementation of Open OnDemand, visit the [HUIT Open OnDemand documentation page](#). If your question can't be answered there, check with your instructor or other course staff, and write to ithelp@harvard.edu if you can't figure it out together.

ARE YOU ABLE TO REACH THE CLUSTER DASHBOARD?

[Click To Answer](#)

ACCESSING THE CLUSTER: FILE SYSTEM



Updates to HUIT OOD configuration

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ACCESSING THE CLUSTER: SHELL ACCESS

The screenshot shows a web browser window with the URL `ood.huit.harvard.edu/pun/sys/dashboard`. The browser's address bar and tabs are visible. Below the browser, the Open OnDemand dashboard is shown. The top navigation bar includes links for 'Open OnDemand', 'Files', 'Jobs', 'Clusters', and 'Interactive Apps'. The 'Clusters' link is highlighted with a yellow tooltip that reads '>_academic Shell Access'. On the left side of the dashboard, the 'OPEN OnDemand' logo is visible, along with sections for 'Latest News' and 'Updates to the HUIT OOD'. A prominent pink warning box with a lightning bolt icon is overlaid on the dashboard. The warning text states: 'Login nodes are not for computing! Login nodes are shared among many users and therefore should not be used to run computationally intensive tasks (R, python, etc). Those tasks should be submitted to the slurm scheduler which will dispatch them on compute nodes.' To the right of the warning box, there is a line of text that reads: 'jobs is expected until includes increased'.

Dashboard - Open OnDemand

ood.huit.harvard.edu/pun/sys/dashboard

CS-2050 Harvard AI Sandbox

Open OnDemand Files Jobs Clusters Interactive Apps

>_academic Shell Access

OPEN OnDemand

OnDemand

Latest News

Updates to the HUIT OOD

The HUIT OOD 5pm. This update includes increased compute resources.

⚡ Login nodes are not for computing!

Login nodes are shared among many users and therefore should not be used to run computationally intensive tasks (R, python, etc). Those tasks should be submitted to the slurm scheduler which will dispatch them on compute nodes.

jobs is expected until includes increased

ACCESSING THE CLUSTER: SHELL ACCESS

```
Last login: Wed Jan 29 23:45:17 2025 from ip-10-37-33-205.ec2.internal
```

```
*****
```

```
*
```

```
*
```

```
*           Welcome to HUIT Open OnDemand!
```

```
*
```

```
*
```

```
*
```

```
* This login node is for lightweight tasks only.
```

```
*
```

```
*
```

```
*
```

```
* For intensive work, use Slurm commands like
```

```
*
```

```
* sbatch or srun to submit jobs to the batch queue.
```

```
*
```

```
* This ensures fair resource allocation and prevents
```

```
*
```

```
* server overload.
```

```
*
```

```
*
```

```
*
```

```
* For more information on using Slurm, type:
```

```
*
```

```
* $ man sbatch or $ man srun
```

```
*
```

```
*
```

```
*
```

```
*****
```

```
[wcw398@ip-10-37-33-44 ~]$ echo "Hello, CS-2050."
```

```
Hello, CS-2050.
```


ACCESSING THE CLUSTER: SHELL ACCESS

```
[wcw398@ip-10-37-33-44 ~]$ echo "Hello, CS-2050."  
Hello, CS-2050.
```

a simple hello

ARE YOU ABLE TO RUN COMMANDS FROM THE LOGIN NODE?

[Click To Answer](#)

GETTING TO KNOW THE CLUSTER

```
[ $ ] lscpu
```

```
[wcw398@ip-10-37-33-28 ~]$ lscpu
Architecture:          x86_64
  CPU op-mode(s):      32-bit, 64-bit
  Address sizes:        46 bits physical, 48 bits virtual
  Byte Order:           Little Endian
CPU(s):                 8
  On-line CPU(s) list: 0-7
Vendor ID:              GenuineIntel
  Model name:           Intel(R) Xeon(R) Platinum 8275CL CPU @ 3.00GHz
    CPU family:         6
    Model:              85
    Thread(s) per core: 2
    Core(s) per socket: 4
    Socket(s):          1
```

COPY EXAMPLES

- Copy the lecture-02 examples to your home directory

```
[$]  cp -r ~/161588/lecture-02 ~  
[$]  chmod u+w --recursive ~/lecture-02
```

- Change to the first example

```
[$]  cd ~/lecture-02/example-1
```

ARE YOU ABLE TO REACH THE FIRST EXAMPLE?

[Click To Answer](#)

SLURM BASICS

sinfo
sbatch
squeue
scancel
srun

Provides basic information
about the cluster

```
[$] sinfo
```

SLURM BASICS

sinfo
sbatch
squeue
scancel
srun

Sends a job (or jobs) to the queue

```
[ $ ] sbatch sbatch-me.sh
```

Be sure to inspect sbatch-me.sh

SLURM BASICS

sinfo
sbatch
squeue
scancel
srun

Replace with your NetID



```
[ $ ] squeue -u wcw398
```

SLURM BASICS

sinfo
sbatch
squeue
scancel
srun

Replace with one of your jobs



```
[$]  scancel 33524
```

```
[wcw398@ip-10-37-33-44 tools]$ squeue -u wcw398
JOBID PARTITION NAME USER ST TIME NODES NODELIST(REASON)
33524 general my_job_n wcw398 R 12:24 1 general-dy-general-cr-5
```

SLURM BASICS

sinfo
sbatch
squeue
scancel
srun

srun has different meanings in different contexts

The command below allows us to work interactively on a compute node

```
[$] srun --nodes=1 --ntasks=4 --time=02:00:00 --pty bash
```

HAVE YOU REACHED A COMPUTE NODE?

[Click To Answer](#)

EXAMPLE 2

```
[$] cd ~/lecture-02/example-2
```

EXAMPLE 2

$$\pi = 2 \int_{-1}^1 \sqrt{1 - x^2} \, dx$$

- Compiling simple programs is easy

```
[ $\$$ ] g++ code.cpp
```

- Compiling modern software can be hard
 - Potentially many libraries and other dependencies
 - Linux/Mac/Windows



CMAKE: TWO STEPS

- Configure

```
[ $\$$ ] cmake -B where-to-build .
```

What does this dot mean?



- Build

```
[ $\$$ ] cmake --build where-to-build
```

HAVE YOU COMPLETED EXAMPLE 2?

[Click To Answer](#)

EXAMPLE 3

$$\pi = 2 \int_{-1}^1 \sqrt{1 - x^2} \, dx$$

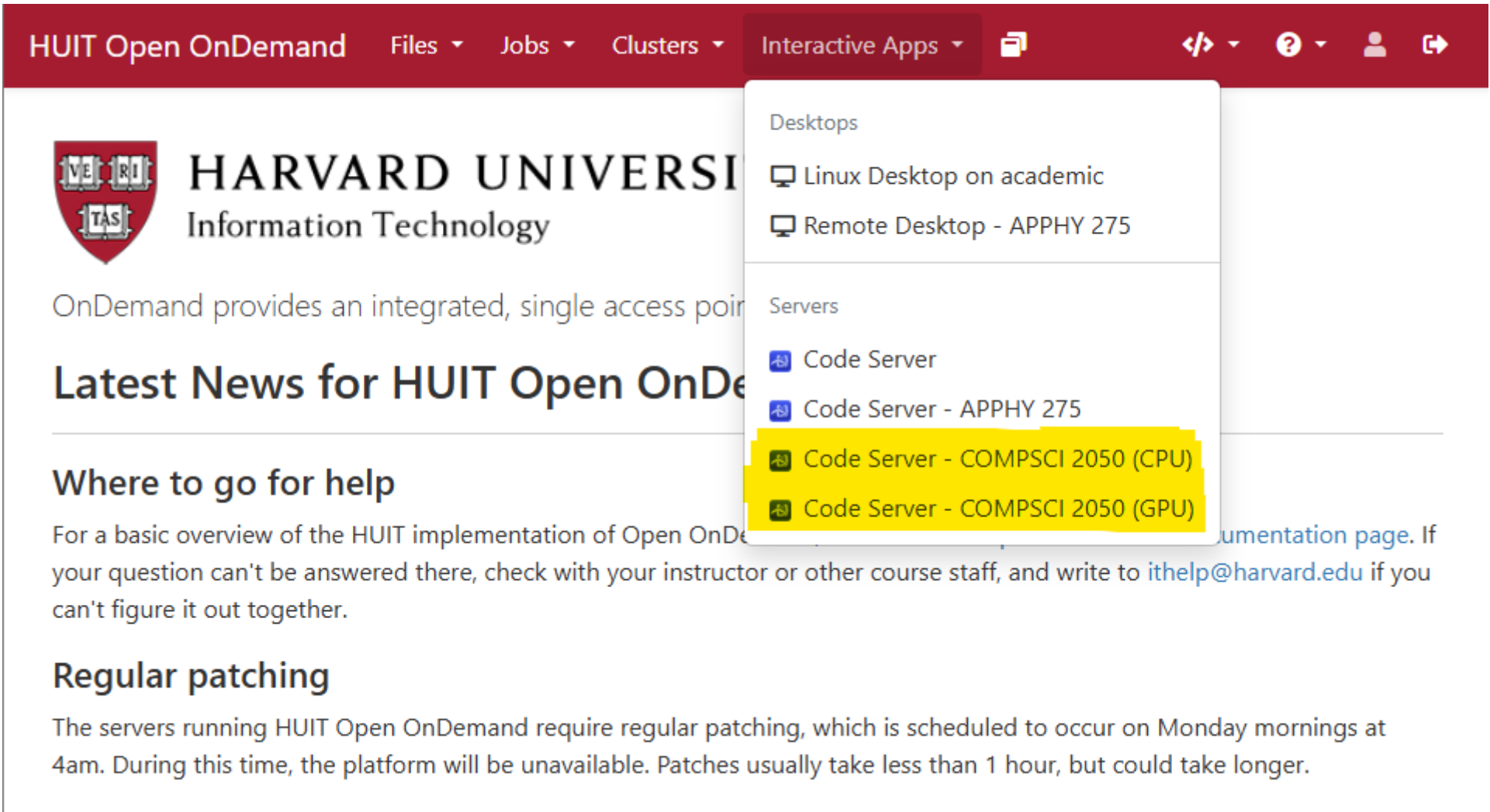
- Python is convenient, but sometimes it is slow
- Modern applications often use a hybrid approach
 - Python-based backbone and tests
 - Computationally intensive parts accelerated with compiled languages (C++, CUDA, etc.)

pybind11

HAVE YOU COMPLETED EXAMPLE 3?

[Click To Answer](#)

ACCESSING THE CLUSTER: VS CODE



The screenshot shows the HUIT Open OnDemand web interface. The top navigation bar is dark red with white text and icons. The 'Interactive Apps' menu is open, displaying a list of available desktops and servers. The background content includes the Harvard University logo and text about OnDemand services, help resources, and server patching schedules.

HUIT Open OnDemand Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾

HARVARD UNIVERSITY
Information Technology

OnDemand provides an integrated, single access point for Harvard's research computing resources.

Latest News for HUIT Open OnDemand

Where to go for help

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Regular patching

The servers running HUIT Open OnDemand require regular patching, which is scheduled to occur on Monday mornings at 4am. During this time, the platform will be unavailable. Patches usually take less than 1 hour, but could take longer.

Desktops

- Linux Desktop on academic
- Remote Desktop - APPHY 275

Servers

- Code Server
- Code Server - APPHY 275
- Code Server - COMPSCI 2050 (CPU)
- Code Server - COMPSCI 2050 (GPU)

HAVE YOU CONNECTED TO VS CODE SUCCESSFULLY?

[Click To Answer](#)



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High Performance Computing for Science and Engineering

CS 2050 and CSCI E-205

questions-and-answers

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main

Q

Go to file

t

+

<> Code

wcw398

update readme

4be70a2 · 17 hours ago

4 Commits

README.md

update readme

17 hours ago

BOOKMARKS

README

Questions and Answers

Please create issues in this repository to ask questions about the course.

CS-2050 / questions-and-answers

Q

Type / to search

+

<> Code

Issues

Pull requests

Actions

Projects

Wiki

Security

Insights

Settings

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Q

Go to file

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About

About

Ask your CS 2050 questions here.

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- Activity
- Custom properties
- 0 stars
- 0 watching
- 0 forks

Releases

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